

Optimal Designs: D, A, I

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Introduction

Computer-generated optimal designs minimise a chosen criterion on the information matrix. When a standard design does not fit constraints (irregular region, mixed-level factors, restricted runs), optimal designs are the primary recourse.

Prerequisites

Linear regression; Fisher information matrix.

Theory

For a design matrix X under the assumed model, the information matrix is $M = X^T X$. Common criteria:

- **D-optimal**: maximise $\det(M)$ – minimises the volume of the joint confidence ellipsoid for coefficients.
- **A-optimal**: minimise $\text{trace}(M^{-1})$ – minimises the average variance of coefficient estimates.
- **I-optimal**: minimise average prediction variance over the design region.
- **G-optimal**: minimise the maximum prediction variance.

Computer algorithms (Fedorov exchange) search for the optimum.

Assumptions

Model is correctly specified; $n \geq p$ (parameters); design region is explicit.

R Implementation

```
library(AlgDesign)

set.seed(2026)
# Candidate set: 3^2 factorial grid
candidates <- expand.grid(x1 = c(-1, 0, 1), x2 = c(-1, 0, 1))

# D-optimal design for a second-order model, selecting 8 runs
d_opt <- optFedorov(~ quad(x1, x2),
                    data = candidates,
                    nTrials = 8,
                    criterion = "D",
                    nRepeats = 20)

d_opt$design

# I-optimal
i_opt <- optFedorov(~ quad(x1, x2),
                    data = candidates,
                    nTrials = 8,
                    criterion = "I",
```

```
i_opt$design  
nRepeats = 20)
```

Output & Results

Two sets of 8 runs from the 9-point candidate set; D-optimal maximises determinant of $X^T X$; I-optimal minimises average prediction variance.

Interpretation

“D-optimal selected the 8 runs that most tightly constrain the polynomial coefficients; I-optimal picked a different subset that minimises prediction variance across the design region. Choose D for inference on coefficients, I for prediction.”

Practical Tips

- Use D-optimal for parameter-precision goals; I-optimal for prediction-focused studies.
- Generate several random starts (**nRepeats**); optimal-design search is non-convex.
- Verify the final design’s efficiency against standard designs if one is applicable.
- Use optimal designs when standard textbook designs don’t fit constraints.
- Mixed-level optimal designs can handle categorical factors; pass them as factors to **optFederov**.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Observational designs and STROBE
- Randomised controlled trials
- Adaptive, non-inferiority, equivalence trials
- Bench and translational design
- MCAR, MAR, MNAR
- Multiple imputation with mice
- Linear mixed models with lme4
- GLMMs and GEE
- DAGs with dagitty and ggdag
- Propensity scores and IPTW
- G-methods, IV, DiD, RDD; heterogeneity of treatment effect
- SIR and SEIR models with deSolve
- Pre-registration and statistical analysis plans